Redox equilibria

Every battery in your phone or car is a pair of redox half-reactions kept apart so that the electrons must travel through a wire. This topic shows how standard electrode potentials measure the pull of each half-reaction for electrons, how they predict which reactions can happen, and how redox titrations put oxidation and reduction to work in quantitative analysis.

SPEC 16.1 Oxidation and reduction: electrons and oxidation numbers

WHAT THIS SPEC POINT ASKS

Define oxidation and reduction as electron transfer and as changes in oxidation number, and apply this to s-, p- and d-block elements.

The two definitions you use at A level

	In terms of electrons	In terms of oxidation number
Oxidation	**loss** of electrons	oxidation number **increases**
Reduction	**gain** of electrons	oxidation number **decreases**

- Memory aid: **OIL RIG** (Oxidation Is Loss, Reduction Is Gain).
- An **oxidising agent** takes electrons, so it is itself **reduced**. A **reducing agent** gives electrons, so it is itself **oxidised**.
- Oxygen and hydrogen definitions (gain of O, loss of H) still work for simple cases, but only the electron and oxidation-number definitions work for everything.

Across the blocks

Block	Typical behaviour	Example
s-block metals	oxidised to 1+ or 2+ ions	$Na \rightarrow Na^+ + e^-$; $Ca \rightarrow Ca^{2+} + 2e^-$
p-block metals	oxidised to positive ions	$Al \rightarrow Al^{3+} + 3e^-$; Sn shows +2 and +4

Block	Typical behaviour	Example
p-block non-metals	usually reduced to negative ions (charge = group number − 8)	$Cl_2 + 2e^- \rightarrow 2Cl^-$; $O_2 + 4e^- \rightarrow 2O^{2-}$
d-block metals	oxidised to ions in **several** oxidation states	Fe^{2+} and Fe^{3+}; Mn from +2 to +7

High d-block oxidation states rarely exist as bare ions. They are found in **oxo-ions**: V(+5) is VO_2^+, V(+4) is VO^{2+}, Mn(+7) is MnO_4^-, Cr(+6) is $Cr_2O_7^{2-}$.

WORKED EXAMPLE

Worked example: oxidation numbers in Unit 5 ions

MnO_4^-: Mn + 4(−2) = −1 → **Mn = +7**

MnO_4^{2-}: Mn + 4(−2) = −2 → **Mn = +6**

$Cr_2O_7^{2-}$: 2Cr + 7(−2) = −2 → **Cr = +6** (the same in CrO_4^{2-})

VO_2^+: V + 2(−2) = +1 → **V = +5**; **VO^{2+}**: V + (−2) = +2 → **V = +4**

$S_2O_3^{2-}$ (thiosulfate): 2S + 3(−2) = −2 → **S = +2**

$S_4O_6^{2-}$ (tetrathionate): 4S + 6(−2) = −2 → **S = +2.5** (an average)

$C_2O_4^{2-}$ (ethanedioate): 2C + 4(−2) = −2 → **C = +3**

$[Rh(CO)_2I_2]^-$: CO is neutral, each I is −1 → Rh − 2 = −1 → **Rh = +1**

Balancing a half-equation in acid

Use this every time; it works for any oxo-ion.

1. Balance the element that changes oxidation number.
2. Balance **O** by adding **H_2O**.
3. Balance **H** by adding **H^+**.
4. Balance **charge** by adding **electrons** (e^-) to the more positive side.

WORKED EXAMPLE

Worked example: MnO_4^- to Mn^{2+}

Mn already balanced: $MnO_4^- \rightarrow Mn^{2+}$

Add $4H_2O$ for the oxygens: $MnO_4^- \rightarrow Mn^{2+} + 4H_2O$

Add $8H^+$ for the hydrogens: $MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$

Charges: left +7, right +2. Add $5e^-$ to the left:

$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)$

Check: Mn goes +7 → +2, a gain of 5 electrons. ✓

To build the **overall** equation, multiply each half-equation so the electrons are equal, add them, then cancel electrons and any H^+ or H_2O that appear on both sides.

MARK-SCHEME WORDING

- Name the **element** and give both oxidation numbers: "Mn reduced from (+)4 to (+)2"; "Cl oxidised from −1 to 0 (in Cl_2)" (WCH15 Jan 2023).
- DO NOT ACCEPT "HNO_3 is reduced": say "nitrogen is reduced from +5 to +2" (WCH15 Jun 2023).
- Not redox: "the oxidation number of vanadium is +5 in both species". IGNORE "there are no electrons in the equation" (WCH15 Oct 2020).

Often asked as: which element is oxidised/reduced and how you know; is this reaction redox?; oxidation number of a metal in an unfamiliar complex.

COMMON MISTAKE

Naming the compound or ion ("nitric acid is reduced") instead of the element.

Saying H or O is oxidised or reduced when it is really the metal that changes.

Calling a reaction redox just because the formulae change: $Cr_2O_7^{2-} \rightarrow CrO_4^{2-}$ is **not** redox (Cr is +6 in both).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Using the **change** in oxidation number from titration data to find an unknown final oxidation state (WCH15 Oct 2021).

Oxidation numbers in organometallic catalysts, e.g. Rh is +1 in $[Rh(CO)_2I_2]^-$ (WCH15 Jan 2026).

SPEC 16.2 Standard electrode potential, E°

WHAT THIS SPEC POINT ASKS

Know what a standard electrode potential is.

What it measures

- A **half-cell** is an element in contact with its ions (or two ions of one element), set up so an equilibrium like $Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ is established at the electrode surface.
- You can't measure the potential of one half-cell on its own. You can only measure the **potential difference** between two half-cells.
- So every half-cell is compared with the same reference: the **standard hydrogen electrode** (SHE), defined as **0.00 V**.

KEY IDEA

The **standard electrode potential**, E°, of a half-cell is the **potential difference (emf)** of a cell made from that half-cell and a **standard hydrogen electrode**, measured under **standard conditions** (298 K, 1.00 mol dm^{-3} ions, 100 kPa gases).

What the sign tells you

- Half-equations are always written as **reductions** (electrons on the left).
- **More positive E°**: the left-hand species is **more easily reduced** than H^+. It is a better **oxidising agent**. Example: $Cl_2 + 2e^- \rightleftharpoons 2Cl^-$, E° = +1.36 V.
- **More negative E°**: the right-hand species is **more easily oxidised**. It is a better **reducing agent**. Example: $Zn^{2+} + 2e^- \rightleftharpoons Zn$, E° = −0.76 V, so zinc readily gives electrons.

EXAM TIP

E° is never asked as a bare definition in WCH15 papers so far. It is tested through the conditions and the SHE inside a "draw the cell" or "describe how to measure" question. Learn the definition anyway: it is the backbone of those answers.

COMMON MISTAKE

Writing a half-equation as an oxidation when quoting E°. Data-book half-equations are always reductions.

SPEC 16.3 Standard conditions for E°

WHAT THIS SPEC POINT ASKS

Know the standard conditions for measuring E°: 298 K, 100 kPa for gases, 1.00 mol dm^{-3} for ions.

Condition	Value	Why it matters
Temperature	**298 K** (25 °C)	E depends on temperature
Pressure of any gas	**100 kPa**	changes the position of an equilibrium involving a gas
Concentration of each ion	**1.00 mol dm^{-3}**	changes the position of the half-cell equilibrium

- In a half-cell with two ions (e.g. Fe^{3+}/Fe^{2+}), **each** ion is at 1.00 mol dm^{-3}.
- If H^+ appears in the half-equation (MnO_4^-, $Cr_2O_7^{2-}$), $[H^+]$ must also be 1.00 mol dm^{-3}.
- A **high-resistance voltmeter** is used so almost no current flows. If current flowed, the concentrations would change and the reading would drift.

MARK-SCHEME WORDING

- "1 mol dm^{-3}" for each ion (stated once if all the same); "hydrogen gas at 100 kPa"; "298 K / 25 °C". The temperature "may be shown anywhere" on a diagram, but it must be there (WCH15 Jun 2023).
- ACCEPT 101 kPa, 1 atm, 1 bar or 1 × 10^5 Pa for the pressure.

Often asked as: the conditions labelled on a cell diagram; MCQ on the pressure in Pa (100 000 Pa).

COMMON MISTAKE

Leaving out the temperature because it is not "attached" to a piece of apparatus (WCH15 Jun 2023 examiner report). Forgetting the H_2 pressure or the H^+ concentration (WCH15 Oct 2023 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Dilution on mixing. If you mix equal volumes of two solutions to make the half-cell electrolyte, each concentration halves. $Fe_2(SO_4)_3$ supplies **two** Fe^{3+} per formula unit, so 0.5 mol dm^{-3} $Fe_2(SO_4)_3$ gives 1.0 mol dm^{-3} Fe^{3+} (WCH15 Jun 2021).

SPEC 16.4 The standard hydrogen electrode and why a reference is needed

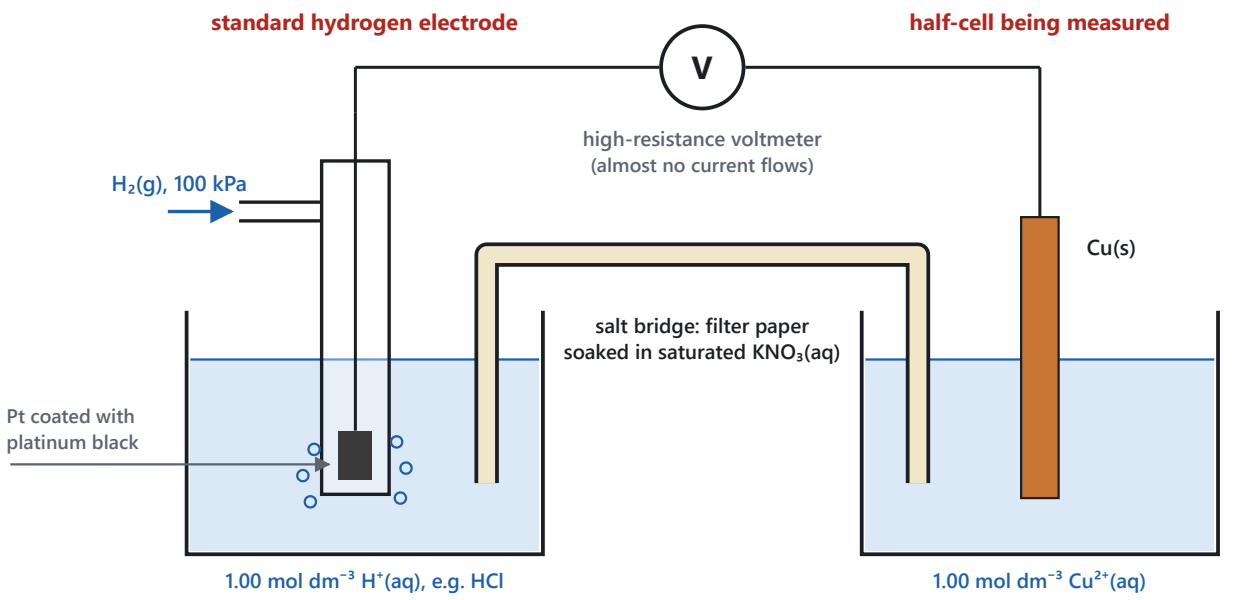
WHAT THIS SPEC POINT ASKS

Know the features of the standard hydrogen electrode (SHE) and why a reference electrode is needed.

Why you need a reference

- A voltmeter needs **two** connections. Putting a second wire into the same solution would create a second electrode with its own potential, so a single electrode potential can never be measured.
- The fix: agree on one half-cell as the zero and measure everything against it. That reference is the SHE, given a potential of **exactly 0.00 V**.

Features of the SHE



Measuring E° for Cu^{2+}/Cu. The SHE (left) is the reference; the voltmeter reads +0.34 V, with the copper electrode positive.

- **Hydrogen gas at 100 kPa** bubbled over the electrode.
- A solution with **$[H^+] = 1.00$ mol dm^{-3}** (e.g. 1.00 mol dm^{-3} HCl).
- An inert **platinum** electrode coated with **platinum black** (finely divided Pt). The large surface area lets the equilibrium $2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ be reached **quickly**, and platinum conducts electrons without reacting.
- **298 K**.

Why it's awkward in practice

The SHE needs a steady supply of hydrogen at exactly 100 kPa, the platinum is easily "poisoned" by impurities, and equilibrium is slow to set up. Labs often use a **secondary standard** (such as a calomel electrode) whose potential against the SHE is known.

MARK-SCHEME WORDING

- "1 mol dm^{-3} $H^+(aq)$ and platinum (black) electrode"; "hydrogen gas … 100 kPa (at 298 K)"; "hydrogen gas must be able to enter the tube" (WCH15 Oct 2023).
- Platinum black is there "to increase the rate of the equilibrium between hydrogen gas and hydrogen ions" (WCH15 Jan 2021).
- Advantage of a calomel electrode (any one): "does not require a continuous supply of hydrogen"; "platinum is easily poisoned"; "difficult to ensure the hydrogen electrode is at equilibrium". IGNORE cheaper, safer, quicker, "hydrogen is explosive" (WCH15 Jun 2023).

Often asked as: draw and label the SHE as part of a cell; purpose of platinum black; advantage of another reference electrode.

COMMON MISTAKE

Hydrogen drawn with no way into the tube, or flowing away from the platinum; one beaker with no liquid; a salt bridge that doesn't dip into both solutions (WCH15 Oct 2023 examiner report).

Using 1 mol dm^{-3} H_2SO_4: that gives 2 mol dm^{-3} H^+. Use 1 mol dm^{-3} HCl. (0.5 mol dm^{-3} H_2SO_4 gives 1 mol dm^{-3} H^+; one mark scheme refused it, two later ones accepted it, so HCl is the safe answer.)

⚠ Beyond the spec · not part of the Edexcel IAL specification

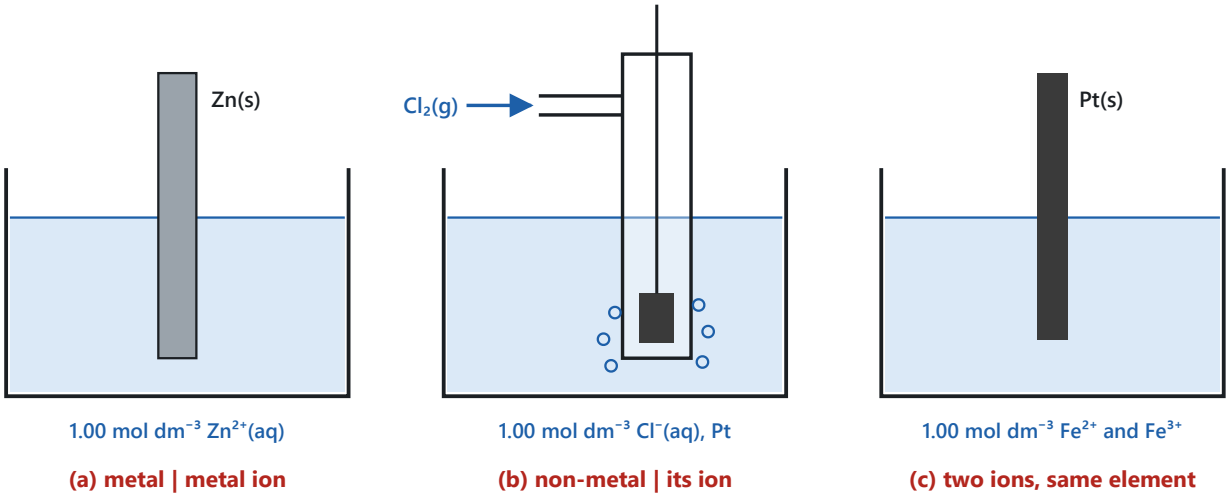
Examined in past papers even though the spec doesn't list it.

Calomel electrode (Hg_2Cl_2 | Hg in saturated KCl), E = +0.24 V against the SHE. You are given its value and work out the other electrode from the measured emf. The KCl is saturated to keep **$[Cl^-]$ constant** (not "to act as a salt bridge") (WCH15 Jun 2023).

SPEC 16.5 Measuring E° for different kinds of half-cell

WHAT THIS SPEC POINT ASKS

Understand the different set-ups for (i) a metal or non-metal in contact with its ions and (ii) two ions of the same element in different oxidation states.



Each half-cell is joined to a SHE by a salt bridge and a high-resistance voltmeter.

Type	Electrode	Solution	Example
Metal / its ion	the **metal itself**	1.00 mol dm^{-3} of its ion	Zn in $Zn^{2+}(aq)$: $Zn^{2+} + 2e^- \rightleftharpoons Zn$, −0.76 V
Non-metal / its ion	**platinum** (the non-metal can't conduct, or is a gas or liquid)	1.00 mol dm^{-3} of the ion; gas at 100 kPa	Cl_2 bubbled over Pt in $Cl^-(aq)$: +1.36 V; Br_2/Br^- (both in solution): +1.09 V
Two ions of one element	**platinum**	**both** ions at 1.00 mol dm^{-3} (plus 1.00 mol dm^{-3} H^+ if H^+ is in the half-equation)	Fe^{3+}/Fe^{2+}: +0.77 V; MnO_4^-/Mn^{2+} with H^+: +1.51 V

- Platinum is **inert**: it doesn't react, it just provides a surface and carries electrons.
- The **salt bridge** completes the circuit by letting **ions** move between the half-cells. A wire can't do this, because it only carries electrons and would set up extra electrode potentials.
- The salt in the bridge must not react with either solution. KNO_3 is the usual choice. KCl is avoided with Ag^+ (AgCl precipitates) and never use something like $KMnO_4$ (it reacts).

MARK-SCHEME WORDING

- Electrode for Fe^{3+}/Fe^{2+}: "platinum / Pt" (ACCEPT platinised platinum, platinum black). DO NOT ACCEPT "iron" (WCH15 Oct 2020).
- Solution: "Fe^{2+} and Fe^{3+} ions, 1 mol dm^{-3} of each", e.g. "1 mol dm^{-3} $FeSO_4$ and 0.5 mol dm^{-3} $Fe_2(SO_4)_3$" (WCH15 Jan 2025).
- Salt bridge: "(saturated) potassium nitrate"; function: "ions can flow through a salt bridge (but not through a wire)". DO NOT ACCEPT "electrons flow through the salt bridge" (WCH15 Jan 2025).

Often asked as: label the parts of a Fe^{3+}/Fe^{2+} half-cell; choose the electrode for an unfamiliar couple; why a salt bridge and not a wire.

COMMON MISTAKE

Giving "Fe^{2+} and Fe^{3+}" with no concentrations or no named compound. Explaining the salt bridge as "balancing charge" without saying **ions move** (WCH15 Jan 2025 examiner report).

Using a metal electrode for a couple of two ions: the Fe^{3+}/Fe^{2+} half-cell needs Pt, not iron.

SPEC 16.6 Core Practical 12: investigating electrochemical cells

WHAT THIS SPEC POINT ASKS

Set up electrochemical cells, measure their emf, and compare the results with values calculated from standard electrode potentials.

CORE PRACTICAL

Core practical 12: method in brief

1. Clean strips of metal (e.g. Cu, Zn, Fe, Ag) with **emery paper** to remove the oxide layer, then rinse.
2. Put each strip in a small beaker of a **1.0 mol dm^{-3}** solution of its own ions (e.g. $CuSO_4$, $ZnSO_4$, $FeSO_4$; $AgNO_3$ for silver).
3. Make a salt bridge from a strip of **filter paper soaked in saturated KNO_3(aq)**, dipping into both beakers.
4. Connect the two metals to a **high-resistance voltmeter** with leads and crocodile clips. Record the reading once it is steady.
5. Repeat for other pairs, using a fresh salt bridge each time.
6. For a two-ion half-cell, use a **platinum** (or carbon) electrode in a solution containing both ions, e.g. Fe^{2+} and Fe^{3+} at 1.0 mol dm^{-3} each.
7. Compare each measured emf with E°cell calculated from the Data Booklet.

Making sense of the results

- The metal connected to the **negative** terminal (when the reading is positive) is the half-cell with the more negative E°: it is where oxidation happens.
- Measured values are usually a little **lower** than the Data Booklet values, because:

- conditions are not standard (room temperature isn't exactly 298 K; concentrations aren't exactly 1.00 mol dm^{-3}); - oxide layers or impurities remain on the metal surface; - the voltmeter draws a small current.

- School labs often use Cu^{2+}/Cu (+0.34 V) as the reference instead of a SHE, then add 0.34 V to convert.

Safety

$AgNO_3$ stains skin and is an irritant; $CuSO_4$ is harmful and toxic to aquatic life; wear eye protection and gloves where needed. Dispose of metal-ion solutions in the residues container, not the sink.

MARK-SCHEME WORDING

For "describe how you would measure E° of $Co^{2+}|Co$" (6-mark, WCH15 Oct 2024), the indicative points are:

- SHE: H_2 gas, platinum electrode, "1 atm, 1.0 mol dm^{-3} HCl and 298 K";
- salt bridge "filter paper soaked in (saturated) KNO_3";
- "1.0 mol dm^{-3} of a named cobalt salt, e.g. $Co(NO_3)_2$", with a cobalt electrode;
- "(high-resistance) voltmeter … to complete the circuit";
- "voltmeter reading = standard electrode potential".

Often asked as: draw and label the apparatus (3–5 marks); 6-mark description of measuring E° for a metal or for Fe^{3+}/Fe^{2+}.

COMMON MISTAKE

Writing "Co^{2+}(aq), 1.0 mol dm^{-3}" without naming a cobalt **compound** (WCH15 Oct 2024 examiner report).

Using platinum when the question gives you copper and zinc electrodes (WCH15 Jun 2022 examiner report). Drawing the salt bridge as an unlabelled line: label it and say what is in it.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Reading the sign of the voltmeter. The right-hand electrode is connected to the positive terminal, so a positive reading means the left electrode is the negative one (WCH15 Jun 2023).

SPEC 16.7 Calculating E°cell

WHAT THIS SPEC POINT ASKS

Calculate the standard emf of a cell, E°cell, from two standard electrode potentials.

KEY IDEA

E°cell = E°(right) − E°(left) = E°(reduction half-cell) − E°(oxidation half-cell)

For a working (feasible) cell, this is **more positive minus less positive**, so E°cell is positive.

- The half-cell with the **more positive** E° is where **reduction** happens: it is the **positive** terminal.
- The half-cell with the **more negative** E° is where **oxidation** happens: it is the **negative** terminal.
- Never multiply an E° value when you multiply a half-equation to balance electrons. E° is a property of the half-cell, not of the amount.

WORKED EXAMPLE

Worked example: the Daniell cell

$Zn^{2+} + 2e^- \rightleftharpoons Zn$, $E° = -0.76$ V; $Cu^{2+} + 2e^- \rightleftharpoons Cu$, $E° = +0.34$ V

Cu^{2+}/Cu is more positive, so Cu^{2+} is reduced (right); Zn is oxidised (left).

$E°cell = (+0.34) - (-0.76) =$ **+1.10 V**. Copper is the positive electrode.

WORKED EXAMPLE

Worked example: working back to an unknown E°

A cell is made from $Zn^{2+}|Zn$ (left, −0.76 V) and an unknown half-cell (right). E°cell = +1.56 V.

$E°cell = E°right - E°left \rightarrow 1.56 = E°right - (-0.76) \rightarrow$ **E°right = +0.80 V**.

The Data Booklet shows this is $Ag^+ + e^- \rightleftharpoons Ag$.

MARK-SCHEME WORDING

- "E°cell = E°R − E°L (or most positive − least positive)". The **sign** must be shown: "+1.17 V"; "−1.17 (V) scores 1 mark" (WCH15 Jun 2025).
- Choose half-equations only from the Data Booklet when told to: "No other metals will score" (WCH15 Jan 2025).

Often asked as: one-mark MCQ calculations; find an unknown E° from a measured E°cell; choose a metal that gives the largest E°cell.

COMMON MISTAKE

Doubling E° when you double the half-equation. Dropping the + sign. Comparing two E° values in words when the question asks for E°cell (WCH15 Jun 2024 examiner report).

Using the wrong half-equation for a species that appears in two of them, e.g. thiosulfate/sulfur instead of thiosulfate/tetrathionate (WCH15 Oct 2022 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Using a **numbered table** of unfamiliar half-equations, including alkaline ones, and picking the right pair (WCH15 Jun 2021; Jun 2024). Alkaline cells: Zn/MnO_2 (WCH15 Jan 2023).

A nickel–silver cell

1. Ni^{2+}/Ni is −0.25 V and Ag^+/Ag is +0.80 V. Silver is more positive, so Ag^+ is reduced and nickel is oxidised.
2. $E^{\ominus}_{cell}$ = +0.80 − (−0.25) = **+1.05 V**. Cell diagram: Ni(s) | Ni^{2+}(aq) || Ag^+(aq) | Ag(s).
3. Combine the half-equations so the electrons cancel: $Ag^+ + e^- \rightarrow Ag$ is doubled, giving $Ni + 2Ag^+ \rightarrow Ni^{2+} + 2Ag$.

Never multiply $E^{\ominus}$ values when you double a half-equation: they measure a tendency per electron, not an amount.

SPEC 16.8 Cell diagrams

WHAT THIS SPEC POINT ASKS

Write cell diagrams using the conventional representation of half-cells.

The rules

- A **single vertical line** (|) marks a phase boundary, e.g. between a solid metal and a solution.
- A **double vertical line** (||), or a dashed one, is the salt bridge.
- A **comma** separates two species in the **same** phase.
- **Oxidation** half-cell on the **left**, **reduction** half-cell on the **right** (the two r's go together: Reduction on the Right).
- Electrodes go on the **outside** edges. Pt is written in wherever neither species is a metal.
- Write each side in the **direction it reacts**: on the left, reactant (the reduced form) then product; on the right, reactant (the oxidised form) then product.
- Include **state symbols** and, where the half-equations have them, the **molar ratios**, H^+ and H_2O.

WORKED EXAMPLE

Worked examples: cell diagrams

Zn/Cu: **Zn(s) | Zn^{2+}(aq) || Cu^{2+}(aq) | Cu(s)** E°cell = +1.10 V

SHE with copper: **Pt(s) | H_2(g) | $2H^+$(aq) || Cu^{2+}(aq) | Cu(s)** E°cell = +0.34 V

Fe^{2+}/Fe^{3+} with chlorine: **Pt(s) | Fe^{2+}(aq), Fe^{3+}(aq) || Cl_2(g) | $2Cl^-$(aq) | Pt(s)** E°cell = 1.36 − 0.77 = +0.59 V

Chloride with dichromate: **Pt(s) | $2Cl^-$(aq) | Cl_2(g) || [$Cr_2O_7^{2-}$(aq) + $14H^+$(aq)], [$2Cr^{3+}$(aq) + $7H_2O$(l)] | Pt(s)**

MARK-SCHEME WORDING

- Allowed: a comma or a vertical line between ions in one phase; a dashed line for the salt bridge.
- Penalised (once each): missing molar ratios; a single solid line between the half-cells; electrons written in the diagram (WCH15 Oct 2023).
- Part marks: "1 for the whole diagram reversed"; "1 if Pt missing on both sides but otherwise correct".

Often asked as: pick the correct diagram (MCQ); write the diagram for two given half-equations (2 marks).

COMMON MISTAKE

Leaving out the molar ratios, the H^+ and H_2O, or one of the platinum electrodes. Examiners say it is "very rare for both marks to be awarded" (WCH15 Oct 2023 examiner report).

Writing an overall equation or two half-equations when a cell diagram was asked for.

SPEC 16.9 Why the conditions matter: non-standard electrode potentials

WHAT THIS SPEC POINT ASKS

Understand why the conditions matter when measuring an electrode potential, E.

Concentration: think Le Chatelier

Each half-cell is an equilibrium written as a reduction: oxidised form + ne^- ⇌ reduced form.

- **Increase the oxidised form** (e.g. more Cu^{2+}), or decrease the reduced form → equilibrium shifts **right**, electrons are removed → E becomes **more positive**.
- **Decrease the oxidised form** (e.g. less Zn^{2+}) → equilibrium shifts **left**, electrons are released → E becomes **more negative**.

- **H^+ in the half-equation** (MnO_4^-, $Cr_2O_7^{2-}$): more acid shifts the equilibrium right, so E is **more positive** and the ion is a stronger oxidising agent. That is why these oxidants are used in **acid**.

Change	Effect on E of that half-cell	Effect on E°cell (Zn/Cu cell)
$[Cu^{2+}]$ increased (right-hand cell)	more positive	increases
$[Zn^{2+}]$ decreased (left-hand cell)	more negative	increases
$[Zn^{2+}]$ increased	less negative	decreases

Temperature and pressure

- Temperature changes the position of every equilibrium, so E changes with temperature too.
- For a gas electrode, a higher gas pressure has the same effect as a higher concentration.

MARK-SCHEME WORDING

- Lower $[Zn^{2+}]$ in a Zn/Cu cell: "Ecell would increase / become more positive"; "$Zn^{2+} + 2e^- \rightleftharpoons Zn$ would shift to the left, making the Zn half-cell more negative". Standalone marks (WCH15 Jan 2025).
- Concentrated HCl with dichromate (E°cell = 1.33 − 1.36 = −0.03 V under standard conditions): "increases $[H^+]$ in the first equilibrium (shifting it right) so increases E"; "increases $[Cl^-]$ in the second equilibrium (shifting it left) so decreases E"; "the values are very close, so Ecell becomes positive". Every mark needs **both** the shift and its effect on E (WCH15 Oct 2023).
- The practical reason concentrated HCl is not used to acidify dichromate: it would produce **toxic chlorine** (WCH15 Oct 2024).

Often asked as: effect of changing one concentration on E or Ecell; why a reaction with a slightly negative E°cell can still happen.

COMMON MISTAKE

A bare "the conditions are non-standard" earns nothing. Say **which** equilibrium shifts, **which way**, and what that does to E (WCH15 Jan 2025 examiner report).

Saying a concentration above 1 mol dm^{-3} makes a reduction potential less positive. More of the oxidised form makes it **more** positive (WCH15 Jun 2022 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

The formula is always given. For oxidised form + ne^- ⇌ reduced form:

$E = E^\circ + (RT / nF) \ln([\text{oxidised}] / [\text{reduced}])$

R = 8.31 J K^{-1} mol^{-1}, T = 298 K, F = 96 500 C mol^{-1}, n = electrons transferred. For a metal/metal-ion half-cell, [reduced] (the solid metal) is taken as 1, so $E = E^\circ + (RT/nF) \ln[M^{n+}]$.

Asked in WCH15 Oct 2020, Jun 2022 and Oct 2024.

WORKED EXAMPLE

Worked example: Nernst equation both ways

Find E for $Zn^{2+}|Zn$ with $[Zn^{2+}] = 0.010$ mol dm^{-3} ($E^\circ = -0.76$ V, n = 2).

$RT/nF = (8.31 \times 298) \div (2 \times 96\,500) = 0.01283$ V

$E = -0.76 + 0.01283 \times \ln 0.010 = -0.76 + 0.01283 \times (-4.605) = -0.76 - 0.059 =$ **−0.82 V** (more negative, as Le Chatelier predicts).

Find the concentration when E = −0.79 V.

$\ln[Zn^{2+}] = (-0.79 - (-0.76)) \div 0.01283 = -2.338$

$[Zn^{2+}] = e^{-2.338} =$ **0.097 mol dm^{-3}**

Keep unrounded values in your calculator: one paper refused an answer of 9×10^{-4} when the working gave 8.99×10^{-4} (WCH15 Oct 2024).

SPEC 16.10 Using E° to predict whether a reaction is feasible

WHAT THIS SPEC POINT ASKS

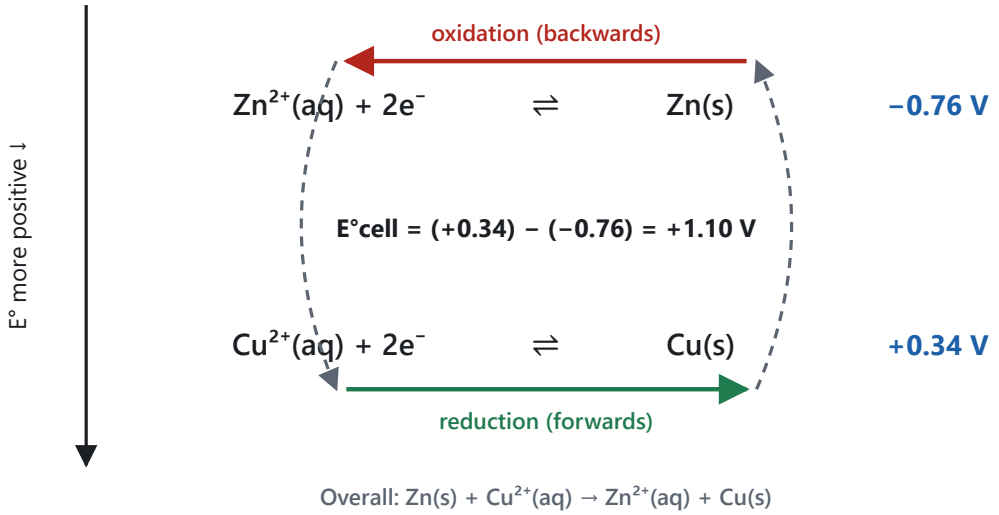
Use standard electrode potentials to predict the thermodynamic feasibility of a reaction.

KEY IDEA

A reaction is **thermodynamically feasible** if **E°cell is positive**, where E°cell = E°(half-cell being reduced) − E°(half-cell being oxidised).

The anticlockwise rule

Write the two half-equations as reductions, the **more negative** one on top. The reaction that goes is **anticlockwise**: the lower (more positive) one goes **forwards** (reduction), the upper one goes **backwards** (oxidation).



Overall: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

The more positive couple is reduced; the more negative couple runs in reverse. Zn reduces Cu^{2+}, not the other way round.

A method that always works

1. Find the two half-equations in the Data Booklet.
2. Decide which species must be **reduced** for the reaction you're asked about.
3. E°cell = E°(that reduction) − E°(the other half-equation).
4. Positive → feasible. Negative → not feasible (the **reverse** reaction is feasible).
5. Write the overall equation: reverse the oxidation half-equation, balance electrons, add, cancel.

WORKED EXAMPLE

Worked example: will iron(III) oxidise iodide? bromide?

$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$, +0.77 V; $I_2 + 2e^- \rightleftharpoons 2I^-$, +0.54 V; $Br_2 + 2e^- \rightleftharpoons 2Br^-$, +1.09 V

Fe^{3+} with I^-: Fe^{3+} reduced, I^- oxidised. E°cell = 0.77 − 0.54 = **+0.23 V** → **feasible**.

$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$

Fe^{3+} with Br^-: E°cell = 0.77 − 1.09 = **−0.32 V** → **not feasible**. (Instead, Br_2 oxidises Fe^{2+}: +0.32 V.)

WORKED EXAMPLE

Worked example: choosing an oxidising agent

Which halogen could oxidise Fe^{2+} to Fe^{3+}? It needs E° more positive than +0.77 V.

Cl_2 (+1.36 V) and Br_2 (+1.09 V) can; I_2 (+0.54 V) can't.

$Cl_2(aq) + 2Fe^{2+}(aq) \rightarrow 2Cl^-(aq) + 2Fe^{3+}(aq)$, E°cell = 1.36 − 0.77 = **+0.59 V**

MARK-SCHEME WORDING

- E°cell must carry a **+ sign or the word "positive"** (WCH15 Oct 2025).
- Equations and E°cell values score **separately**: four correct half-equations score even if not combined (WCH15 Oct 2022).
- "VO^{2+} is not oxidised any further as E°cell is −0.2 V / negative" (WCH15 Oct 2020). Uncancelled electrons, H^+ or H_2O are penalised once.

Often asked as: write the equation, calculate E°cell and say whether it is feasible (3–5 marks); 6-mark on which reducing agents can take vanadium to which oxidation state.

COMMON MISTAKE

Calculating every E°cell but writing no equations, or the reverse. Give **both** (WCH15 Jan 2023 examiner report).

Not saying where a reduction **stops**, e.g. "so V^{3+} is not reduced to V^{2+} because E°cell is negative" (WCH15 Oct 2022 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

A reaction with a small negative E°cell, e.g. −0.01 V, may still happen to some extent if conditions change (WCH15 Oct 2022). The anticlockwise rule with an unfamiliar middle couple, e.g. Fe^{3+}/Fe^{2+} (+0.77 V) between I_2/I^- and $S_2O_8^{2-}/SO_4^{2-}$ (WCH15 Jan 2026).

Can iron(III) ions oxidise iodide? Bromide?

1. Fe^{3+}/Fe^{2+} is +0.77 V and I_2/I^- is +0.54 V. For Fe^{3+} to be reduced and I^- oxidised, $E^{\ominus}_{cell}$ = 0.77 − 0.54 = +0.23 V: **feasible**. $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$.

2. Br_2/Br^- is +1.09 V. For Fe^{3+} to oxidise Br^-, $E^{\ominus}_{cell}$ = 0.77 − 1.09 = −0.32 V: **not feasible**. The reverse is: bromine oxidises Fe^{2+} to Fe^{3+}.

SPEC 16.11 E°cell, total entropy change and the equilibrium constant

WHAT THIS SPEC POINT ASKS

Understand that E°cell is directly proportional to ΔS°total and to ln K for a reaction.

Three ways of saying the same thing

Measure	Feasible when…	Link
E°cell	positive	—
ΔS°total	positive	ΔS°total ∝ E°cell
ln K	positive (K > 1)	ln K ∝ E°cell

- The **bigger** E°cell, the **bigger** ΔS°total and the **bigger** K. A large positive E°cell means the equilibrium lies far to the right: the reaction goes almost to completion.
- A small positive E°cell (a few tenths of a volt) still gives K > 1, but both reactants and products may be present at equilibrium.
- A negative E°cell means ΔS°total is negative and K < 1: the reverse reaction is the feasible one.

EXAM TIP

Think of three gauges of one quantity. If one is positive, the others are too. MCQs often show a straight line through the origin for E°cell against ΔS°total or ln K: that is the "directly proportional" idea.

MARK-SCHEME WORDING

- "E°cell is proportional to ΔS°total"; "so the total entropy change is positive as E°cell is positive"; "a positive ΔS°total means the reaction is (thermodynamically) feasible" (WCH15 Jan 2026).
- ACCEPT ΔS°total = nFE°cell ÷ T; "(R) ln K is proportional to E°cell".

Often asked as: MCQ on the signs of ΔS°total and ln K for a positive E°cell; a 3-mark explanation linking E°cell to feasibility.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

$\Delta S°_{total} = nFE°_{cell} \div T$ and $\ln K = nFE°_{cell} \div RT$ (accepted in mark schemes, but numerical use has not been required).

For the Daniell cell (n = 2, $E°_{cell}$ = +1.10 V): $\Delta S°_{total} = 2 \times 96\,500 \times 1.10 \div 298$ = **+712 J K^{-1} mol^{-1}**; ln K = 85.7, so $K \approx 10^{37}$: the reaction goes to completion.

SPEC 16.12 Limitations of predictions from E°

WHAT THIS SPEC POINT ASKS

Understand why predictions from standard electrode potentials can be wrong: kinetic stability and non-standard conditions.

E° tells you "can it?", never "how fast?"

- A positive $E°_{cell}$ means the reaction is **thermodynamically feasible**. It says **nothing about the rate**.
- If the **activation energy is high**, a feasible reaction may be too slow to see. The reactants are **kinetically stable** (thermodynamically unstable, but kinetically inert).

Feasible but slow	$E°_{cell}$	Why it's slow
$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$	2.01 − 0.54 = +1.47 V	two **negative ions repel**, so high Ea (see 17.31)
$2H_2O_2 \rightarrow 2H_2O + O_2$	1.77 − 0.68 = +1.09 V	high Ea; fast only with a catalyst such as MnO_2
MnO_4^- + $C_2O_4^{2-}$ at room temperature	positive	anions repel; needs heating and Mn^{2+} (see 17.32)

Conditions are rarely standard

- Real concentrations are not 1.00 mol dm^{-3}, temperatures are not 298 K, and many reactions don't happen in aqueous solution at all. Under different conditions E changes (16.9).
- If two E° values are **close** (within about 0.1 V), a modest change in concentration can reverse the prediction.
- E° values describe aqueous ions. They can't predict reactions between solids or gases, or in concentrated solutions.

MARK-SCHEME WORDING

- Feasible but doesn't happen: "the activation energy for the reaction is high" and "the reactants repel (because they are both negatively charged)" (WCH15 Jan 2026); "reaction rate must be slow / activation energy high" (WCH15 Oct 2022).

- Prediction wrong the other way: "the conditions must be non-standard and the E° values are so close" (WCH15 Jun 2022). ACCEPT "high kinetic energy barrier".

Often asked as: explain why a reaction with a positive E°cell doesn't happen; explain why an unexpected reaction does happen.

COMMON MISTAKE

"The reaction is feasible despite a negative E°cell because conditions are non-standard" with no further reasoning (WCH15 Oct 2023 examiner report). Name the equilibrium and the direction it shifts.

Writing only "no reaction" when asked why, with no reference to rate or activation energy (WCH15 Oct 2022 examiner report).

SPEC 16.13 Standard reduction potentials and the electrochemical series

WHAT THIS SPEC POINT ASKS

Know that standard electrode potentials are also called standard reduction potentials and can be listed as an electrochemical series.

- Because every half-equation is written as a **reduction**, E° values are also called **standard reduction potentials**.
- Listing them in order of E° gives the **electrochemical series**.

Half-equation	E° / V	
$Li^+ + e^- \rightleftharpoons Li$	−3.03	← strongest **reducing agent** (Li)
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	−2.37	
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76	
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	−0.44	
$2H^+ + 2e^- \rightleftharpoons H_2$	0.00	reference
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+0.34	
$I_2 + 2e^- \rightleftharpoons 2I^-$	+0.54	
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+0.77	

Half-equation	E° / V	
$Ag^+ + e^- \rightleftharpoons Ag$	+0.80	
$Br_2 + 2e^- \rightleftharpoons 2Br^-$	+1.09	
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33	
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36	
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	+1.51	
$F_2 + 2e^- \rightleftharpoons 2F^-$	+2.87	← strongest **oxidising agent** (F_2)

- **Oxidising agents** are on the **left**; they get stronger **down** the table.
- **Reducing agents** are on the **right**; they get stronger **up** the table.

MARK-SCHEME WORDING

- "reduction" for "what type of potential is this?". IGNORE "redox", "standard electrode potential" (WCH15 Jun 2024).
- "electrochemical (series)" (ACCEPT reduction potential series). "Reactivity series" does not score (WCH15 Jun 2024).

Often asked as: two one-mark recall items.

COMMON MISTAKE

Writing "reactivity series". Under a quarter of candidates got "electrochemical series" (WCH15 Jun 2024 examiner report): it is straight recall of the spec wording.

SPEC 16.14 Disproportionation and E°

WHAT THIS SPEC POINT ASKS

Use standard electrode potentials to predict whether a disproportionation reaction is thermodynamically feasible.

What disproportionation is

Disproportionation: one species is **both oxidised and reduced** in the same reaction, so one element ends up in a higher **and** a lower oxidation state.

- Example: $2Cu^{+}(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$. Copper goes +1 → +2 **and** +1 → 0.
- The reverse, where two oxidation states combine to make one in between, is **comproportionation**.

How to test it with E°

The species appears in **two** half-equations: once on the **left** (being reduced) and once on the **right** (being oxidised).

1. Find the half-equation where the species is **reduced** (it's on the left).
2. Find the half-equation where the species is **formed** (it's on the right); this one runs **backwards** as the oxidation.
3. E°cell = E°(reduction of the species) − E°(the one that forms it).
4. Positive → the species **disproportionates**. Negative → it is **stable** to disproportionation.

WORKED EXAMPLE

Worked example: copper(I) disproportionates

$Cu^{+} + e^{-} \rightleftharpoons Cu$, E° = +0.52 V ($Cu^{+}$ reduced)

$Cu^{2+} + e^{-} \rightleftharpoons Cu^{+}$, E° = +0.15 V (run backwards: Cu^{+} oxidised)

E°cell = 0.52 − 0.15 = **+0.37 V** → **feasible**

$2Cu^{+}(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$

This is why simple Cu^{+} salts dissolved in water give a brown copper precipitate and a blue solution. Cu(I) is only stable in water when it is in a precipitate (CuI) or a complex.

WORKED EXAMPLE

Worked example: iron(II) does not disproportionate

$Fe^{2+} + 2e^{-} \rightleftharpoons Fe$, E° = −0.44 V ($Fe^{2+}$ reduced)

$Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$, E° = +0.77 V (run backwards: Fe^{2+} oxidised)

E°cell = −0.44 − 0.77 = **−1.21 V** → **not feasible**. $3Fe^{2+} \rightarrow Fe + 2Fe^{3+}$ doesn't happen. The reverse ($Fe + 2Fe^{3+} \rightarrow 3Fe^{2+}$, comproportionation) does.

WORKED EXAMPLE

Worked example: manganate(VI) in acid

$MnO_4^{2-} + 4H^{+} + 2e^{-} \rightleftharpoons MnO_2 + 2H_2O$, E° = +2.26 V ($MnO_4^{2-}$ reduced, Mn +6 → +4)

$MnO_4^{-} + e^{-} \rightleftharpoons MnO_4^{2-}$, E° = +0.56 V (run backwards: MnO_4^{2-} oxidised, Mn +6 → +7)

E°cell = 2.26 − 0.56 = **+1.70 V** → **feasible**

Electrons: the reduction takes $2e^-$, the oxidation gives $1e^-$, so double the oxidation:

$2MnO_4^{2-} \rightarrow 2MnO_4^- + 2e^-$

$MnO_4^{2-} + 4H^+ + 2e^- \rightarrow MnO_2 + 2H_2O$

$3MnO_4^{2-}(aq) + 4H^+(aq) \rightarrow 2MnO_4^-(aq) + MnO_2(s) + 2H_2O(l)$

Green manganate(VI) turns purple, with a brown-black solid.

EXAM TIP

Hydrogen peroxide disproportionates too: O goes −1 → −2 (in H_2O) and −1 → 0 (in O_2). E°cell = 1.77 − 0.68 = +1.09 V, so $2H_2O_2 \rightarrow 2H_2O + O_2$ is feasible. It is slow without a catalyst: a link to 16.12.

Latimer-style diagrams

Some questions give the E° values on a line of species from highest to lowest oxidation state, with the E° for each step written over the arrow:

Cu^{2+} —(+0.15 V)→ Cu^+ —(+0.52 V)→ Cu

A species in the middle **disproportionates if the E° on its right is more positive than the E° on its left** (here 0.52 > 0.15, so Cu^+ disproportionates). This is the same calculation as above in a compact form.

MARK-SCHEME WORDING

- "$Cu^{2+} + e^- \rightleftharpoons Cu^+$ E° = +0.15 V and $Cu^+ + e^- \rightleftharpoons$ Cu E° = +0.52 V"; "E°cell = 0.52 − 0.15 = (+)0.37 (V) and therefore the reaction is (thermodynamically) feasible". "No TE on incorrect half-equations" (WCH15 Jan 2021).
- "Unbalanced equation with all species correct scores (1)" (WCH15 Jun 2021).
- Comproportionation with E°cell = −0.03 V: "the E° values are close"; "the reaction may be shifted in the required direction using concentrated alkali" (WCH15 Jun 2021).

Often asked as: write the ionic equation and use E° values to show the disproportionation is feasible; derive the equation for an unfamiliar Mn species from given clues.

COMMON MISTAKE

Choosing the wrong pair of half-equations: you need the two in which the **same** species appears, once on each side (WCH15 Jan 2021 examiner report).

Getting the three Mn species right but failing to balance. Write the two half-equations first, then combine (WCH15 Jan 2024 examiner report).

Not saying the E°cell is **positive** after calculating it.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Disproportionation in **alkali** using the alkaline half-equation: $3MnO_4^{2-} + 2H_2O \rightarrow 2MnO_4^- + MnO_2 + 4OH^-$ (WCH15 Jun 2021).

Unfamiliar species from a Latimer-style table, e.g. H_3MnO_4 (Mn +5) → MnO_4^- + MnO_2, E°cell = +1.62 V (WCH15 Jan 2024).

Spotting a disproportionation in an MCQ: $2Ag^{2+} \rightarrow Ag^+ + Ag^{3+}$ (WCH15 Oct 2021).

SPEC 16.15 Redox titration calculations

WHAT THIS SPEC POINT ASKS

Carry out structured and unstructured redox titration calculations, including iron(II) with manganate(VII) and iodine with thiosulfate.

The two key reactions

Titration	Ionic equation	Mole ratio
Manganate(VII) and iron(II)	$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$	**MnO_4^- : Fe^{2+} = 1 : 5**
Manganate(VII) and ethanedioate	$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$	**MnO_4^- : $C_2O_4^{2-}$ = 2 : 5**
Dichromate(VI) and iron(II)	$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$	**$Cr_2O_7^{2-}$: Fe^{2+} = 1 : 6**
Iodine and thiosulfate	$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$	**I_2 : $S_2O_3^{2-}$ = 1 : 2**

The route through any titration calculation

1. Write (or find) the balanced equation and the mole ratio.
2. **Moles of the titrant** = concentration × titre (in dm^3).
3. Use the ratio to get **moles of the other reactant** in the portion titrated.
4. **Scale up** to the whole sample (e.g. × 10 for 25.0 cm^3 taken from 250.0 cm^3).
5. Convert to what is asked: mass (× M), concentration (÷ volume), percentage.

Iodine–thiosulfate titrations are usually **indirect**: an oxidising agent reacts with **excess** iodide to release I_2, and you titrate the I_2. Chain the ratios: oxidant → I_2 → $S_2O_3^{2-}$.

WORKED EXAMPLE

Worked example: purity of iron(II) sulfate (MnO_4^-, structured)

5.00 g of impure $FeSO_4 \cdot 7H_2O$ (M = 277.9 g mol^{-1}) is dissolved in dilute sulfuric acid and made up to 250.0 cm^3. 25.0 cm^3 portions need a mean titre of 17.60 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$.

$n(MnO_4^-) = 0.0200 \times 17.60/1000 = 3.52 \times 10^{-4}$ mol

$n(Fe^{2+})$ in 25.0 $cm^3 = 5 \times 3.52 \times 10^{-4} = 1.76 \times 10^{-3}$ mol

$n(Fe^{2+})$ in 250.0 $cm^3 = 1.76 \times 10^{-2}$ mol

mass $FeSO_4 \cdot 7H_2O = 1.76 \times 10^{-2} \times 277.9 = 4.891$ g

purity = 4.891 ÷ 5.00 × 100 = **97.8%**

WORKED EXAMPLE

Worked example: copper in brass (iodine–thiosulfate)

0.950 g of brass is dissolved in nitric acid, neutralised and made up to 250.0 cm^3. A 25.0 cm^3 portion is added to excess KI: $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$. The iodine needs 19.00 cm^3 of 0.0500 mol dm^{-3} $Na_2S_2O_3$.

$n(S_2O_3^{2-}) = 0.0500 \times 0.01900 = 9.50 \times 10^{-4}$ mol

$n(I_2) = 9.50 \times 10^{-4} \div 2 = 4.75 \times 10^{-4}$ mol

$n(Cu^{2+}) = 2 \times 4.75 \times 10^{-4} = 9.50 \times 10^{-4}$ mol in 25.0 cm^3 (overall $Cu^{2+} : S_2O_3^{2-} = 1 : 1$)

in 250.0 cm^3: 9.50×10^{-3} mol; mass = $9.50 \times 10^{-3} \times 63.5 = 0.603$ g

% Cu = 0.603 ÷ 0.950 × 100 = **63.5%**

WORKED EXAMPLE

Worked example: an unknown oxidation state (unstructured)

25.0 cm^3 of 0.100 mol dm^{-3} V^{2+} needs 30.0 cm^3 of 0.0500 mol dm^{-3} $KMnO_4$. What is the final oxidation state of vanadium?

$n(MnO_4^-) = 0.0500 \times 0.0300 = 1.50 \times 10^{-3}$ mol; each takes $5e^-$, so electrons accepted = 7.50×10^{-3} mol

$n(V^{2+}) = 0.100 \times 0.0250 = 2.50 \times 10^{-3}$ mol

electrons lost per V = $7.50 \times 10^{-3} \div 2.50 \times 10^{-3} = 3$

V goes from +2 to **+5** (as VO_2^+).

WORKED EXAMPLE

Worked example: a back titration

0.200 g of an ore containing MnO_2 is warmed with 50.0 cm^3 of 0.100 mol dm^{-3} Fe^{2+} in acid (excess): $MnO_2 + 4H^+ + 2Fe^{2+} \rightarrow Mn^{2+} + 2Fe^{3+} + 2H_2O$. The leftover Fe^{2+} needs 20.00 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$.

$n(MnO_4^-) = 0.0200 \times 0.02000 = 4.00 \times 10^{-4}$ mol → leftover $Fe^{2+} = 5 \times 4.00 \times 10^{-4} = 2.00 \times 10^{-3}$ mol

Fe^{2+} added $= 0.100 \times 0.0500 = 5.00 \times 10^{-3}$ mol

Fe^{2+} that reacted with $MnO_2 = 5.00 \times 10^{-3} - 2.00 \times 10^{-3} = 3.00 \times 10^{-3}$ mol

$n(MnO_2) = 3.00 \times 10^{-3} \div 2 = 1.50 \times 10^{-3}$ mol; mass $= 1.50 \times 10^{-3} \times 86.9 = 0.130$ g

% $MnO_2 = 0.130 \div 0.200 \times 100 =$ **65.2%**

MARK-SCHEME WORDING

- Marks are by **process**: "moles of MnO_4^- in the titre"; "use of 1:5 ratio"; "scaling to 250.0 cm^3"; "conversion of moles to g"; "percentage … to 3 SF". "TE at each stage"; "ignore premature rounding if the final answer is correct".
- "Correct answer with no working scores zero" on some papers: always show the route.
- Use Ar from the periodic table in the Data Booklet (55.8 or 56 for Fe both accepted). Atomic numbers used as Ar score nothing (WCH15 Jun 2021).

Often asked as: % of iron in tablets, wire or moss killer (5–6 marks); Cu in a complex salt by iodine–thiosulfate; dissolved oxygen (Winkler); find an unknown oxidation state.

COMMON MISTAKE

Using 25.0 cm^3 where the **titre** belongs, or forgetting the × 10 scale-up (WCH15 Jan 2025 examiner report).

The wrong ratio, or none at all (WCH15 Jan 2024 examiner report). If you get a percentage over 100, **recheck the ratio**: don't fudge it (WCH15 Oct 2020 examiner report).

Rounding intermediate values, and quoting 4 or more SF (WCH15 Oct 2024 examiner report). Using the formula mass of VO_3^- when the question asks for mass of vanadium (WCH15 Oct 2022 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Back titrations: excess reagent added, the leftover titrated, the amount used found by difference (WCH15 Jun 2023; Oct 2022). Many candidates lose marks by missing the back-titration step.

Winkler method for dissolved oxygen: moles of $S_2O_3^{2-} \div 2$ = moles of I_2; $I_2 : MnO_2 = 1 : 1$; $MnO_2 : O_2 = 2 : 1$ (WCH15 Oct 2025).

ppm and ppb. For dilute aqueous solutions 1 $dm^3 \approx 1$ kg, so 1 mg $dm^{-3} \approx 1$ ppm and 1 µg $dm^{-3} \approx 1$ ppb. Example: 2.15×10^{-5} mol dm^{-3} Fe × 55.8 = 1.20×10^{-3} g dm^{-3} = 1.20 mg dm^{-3} = **1.20 ppm** (WCH15 Jan 2025; Jun 2021).

A formula of a hydrate from titration data, e.g. $KCr(SO_4)_2 \cdot 12H_2O$ (WCH15 Jan 2025).

Iron in iron tablets

1. 2.80 g of iron tablets is dissolved in dilute sulfuric acid and made up to 250 cm^3. 25.0 cm^3 portions need 24.60 cm^3 of 0.00500 mol dm^{-3} $KMnO_4$. Find the percentage of iron by mass.
2. Moles of MnO_4^- = 0.02460 × 0.00500 = 1.23×10^{-4} mol.
3. The equation has 5 Fe^{2+} per MnO_4^-: moles of Fe^{2+} in 25.0 cm^3 = $5 \times 1.23 \times 10^{-4} = 6.15 \times 10^{-4}$ mol.
4. In the whole 250 cm^3: 6.15×10^{-3} mol, so the mass of iron = $6.15 \times 10^{-3} \times 55.8 = 0.343$ g.
5. Percentage of iron = 0.343 ÷ 2.80 × 100 = **12.3%**.

Copper in brass

1. 1.50 g of brass is dissolved in nitric acid, neutralised and made up to 250 cm^3. A 25.0 cm^3 portion is added to excess KI, and the iodine released needs 15.70 cm^3 of 0.100 mol dm^{-3} sodium thiosulfate. Find the percentage of copper.
2. Moles of $S_2O_3^{2-}$ = 0.01570 × 0.100 = 1.57×10^{-3} mol.
3. $2Cu^{2+}$ make 1 I_2, which reacts with 2 $S_2O_3^{2-}$, so $Cu^{2+} : S_2O_3^{2-} = 1 : 1$. Moles of Cu^{2+} in 25.0 cm^3 = 1.57×10^{-3} mol.
4. In 250 cm^3: 1.57×10^{-2} mol, so mass of copper = $1.57 \times 10^{-2} \times 63.5 = 0.997$ g.
5. Percentage of copper = 0.997 ÷ 1.50 × 100 = **66.5%**.

SPEC 16.16 Uncertainty in titration results

WHAT THIS SPEC POINT ASKS

Discuss the uncertainty of measurements and what it means for how valid the final result is.

Percentage uncertainty

% uncertainty = (uncertainty ÷ measured value) × 100

Apparatus	Typical uncertainty	Readings	Notes
Burette	±0.05 cm³ per reading	**2** (start and end)	titre uncertainty = ±0.10 cm³
Pipette (25.0 cm³)	±0.06 cm³	1	
Volumetric flask (250.0 cm³)	±0.3 cm³	1	
Balance (2 d.p.)	±0.005 g per reading	**2** if weighing by difference	

- A **bigger titre** gives a **smaller** % uncertainty: ±0.10 on 5.00 cm³ is 2%, but on 25.00 cm³ it is only 0.4%. Choose concentrations so the titre is large (ideally 20–30 cm³).
- Add the % uncertainties of each measurement to get the total % uncertainty of the result.
- **% uncertainty is not % error.** Error compares your result with an accepted value.

WORKED EXAMPLE

Worked example: is 97.8% justified?

From the $FeSO_4$ titration in 16.15:

burette: 0.10 ÷ 17.60 × 100 = 0.57%

pipette: 0.06 ÷ 25.0 × 100 = 0.24%

flask: 0.3 ÷ 250.0 × 100 = 0.12%

balance: 2 × 0.005 ÷ 5.00 × 100 = 0.20%

total ≈ **1.1%** → 97.8% ± 1.1%, so the result lies between about 96.7% and 98.9%.

The third significant figure is uncertain: quote **98%** (2 SF) or 97.8% (3 SF), never 97.82%.

MARK-SCHEME WORDING

- "100 × 0.05 × 2 ÷ 27.35 = 0.37(%)" for the burette; "100 × 0.06 ÷ 25 = 0.24(%)" for the pipette (WCH15 Jun 2021).
- "the total percentage uncertainty is much bigger than the rounding, so the answer should be to no more than 2 / 3 SF". IGNORE "data given to 3 SF".

Often asked as: calculate % uncertainties for several pieces of apparatus; decide how many SF are sensible; effect of an impurity or a damp sample on the result.

COMMON MISTAKE

Forgetting that a burette (and weighing by difference) involves **two** readings.

Too many significant figures: the examiners' rule is one fewer than the least precise datum, or the same number (WCH15 Jan 2021 examiner report).

SPEC 16.17 Core Practicals 13a and 13b: redox titrations

WHAT THIS SPEC POINT ASKS

Carry out redox titrations with (i) iron(II) and potassium manganate(VII) and (ii) sodium thiosulfate and iodine.

CORE PRACTICAL

Core practical 13a: iron(II) with potassium manganate(VII)

1. Weigh the iron(II) sample accurately (e.g. ground iron tablets or ammonium iron(II) sulfate) by difference.
2. Dissolve in about 100 cm^3 of **dilute sulfuric acid**, transfer with washings to a 250.0 cm^3 volumetric flask, make up to the mark with distilled water and invert to mix.
3. Pipette 25.0 cm^3 into a conical flask and add about 10 cm^3 more dilute H_2SO_4 (excess acid).
4. Fill the burette with $KMnO_4(aq)$ (≈0.02 mol dm^{-3}). Read the **top** of the meniscus, because the purple solution is too dark to see the bottom.
5. Titrate until the first **permanent pale pink** colour. No indicator is needed: MnO_4^- is **self-indicating**.
6. Do a rough titre, then accurate titres until two are **concordant** (within 0.10 cm^3). Average the concordant titres.

CORE PRACTICAL

Core practical 13b: iodine with sodium thiosulfate

1. Pipette 25.0 cm^3 of the oxidising agent (e.g. KIO_3 standard, bleach, or a Cu^{2+} solution) into a conical flask.

2. Add **excess** potassium iodide (and acid where needed). Iodine is released: the solution turns brown.
3. Titrate with $Na_2S_2O_3$(aq) from the burette until the brown fades to **pale straw yellow**.
4. Add about 1 cm^3 of **starch** solution: it turns **blue-black**.
5. Continue dropwise until the blue-black colour **just disappears** (colourless; with copper, an off-white CuI precipitate remains).
6. Repeat to concordant titres.

Why each step matters

Step	Reason
Acidify MnO_4^- with **dilute H_2SO_4**	MnO_4^- needs H^+ to be reduced fully to Mn^{2+}. Too little acid gives a brown MnO_2 precipitate and the ratio changes.
Not HCl	MnO_4^- (+1.51 V) oxidises Cl^- to Cl_2 (+1.36 V), so extra MnO_4^- is used and Cl_2 is toxic.
Not nitric acid	It is itself an oxidising agent and would oxidise Fe^{2+}.
Excess KI	ensures all the oxidant reacts, so moles of I_2 truly reflect the oxidant; it also keeps I_2 dissolved (as I_3^-).
Starch added **near the end**	added too early, a starch–iodine complex forms that breaks up only slowly, so the end point is late and unclear.

MARK-SCHEME WORDING

- MnO_4^- in the burette: "colourless (or pale green) to (pale) pink". MnO_4^- in the flask: "(pale) pink to colourless". DO NOT ACCEPT purple, magenta or "permanent" alone (WCH15 Jan 2024; Jun 2021).
- Iodine–thiosulfate: "blue-black to colourless" (WCH15 Jan 2022).
- Self-indicating: advantage "no indicator needed"; disadvantage "the intense colour of manganate(VII) makes reading the burette difficult" (WCH15 Jan 2026).
- Dichromate needs an indicator because "the colour change (orange to green) is not distinctive / sharp enough". $Cr_2O_7^{2-}$ orange, Cr^{3+} green; DO NOT ACCEPT blue (WCH15 Jun 2022).

Often asked as: the end-point colour change; why KI must be in excess; advantage/disadvantage of a self-indicating titrant.

COMMON MISTAKE

The colour change the wrong way round (it depends on which solution is in the burette), or giving the colours of acid–base indicators (WCH15 Jan 2024 examiner report).

Leaving out the word "colour" when asked why an indicator is needed (WCH15 Jun 2022 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

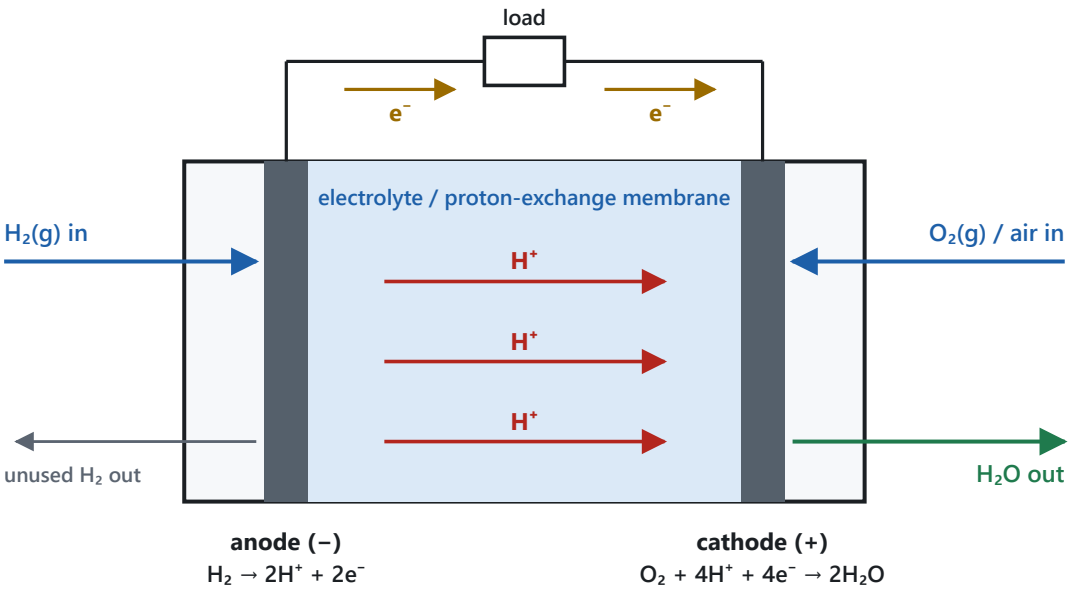
Titrations with **Ce^{4+}** or **Ti^{3+}** as titrant, and heating the flask "to speed up the reaction" (WCH15 Oct 2021; Jan 2026).

SPEC 16.18 Fuel cells

WHAT THIS SPEC POINT ASKS

Understand that fuel cells use the energy released when a fuel reacts with oxygen to generate a voltage; methanol and other hydrogen-rich fuels can be used.

How a fuel cell works



Acid (proton-exchange membrane) fuel cell. H^+ ions move through the electrolyte; electrons move through the external circuit.

- The **fuel** (H_2, methanol, ethanol…) is **oxidised** at the **anode** (negative electrode). Electrons flow round the external circuit.
- **Oxygen** is **reduced** at the **cathode** (positive electrode).
- The overall reaction is the same as burning the fuel, but the energy comes out as **electricity** instead of heat, so no flame is needed.

- Fuel and oxygen are supplied **continuously**, so the cell keeps working as long as they are fed in. It doesn't need recharging, and it doesn't store energy.
- Electrodes are porous and coated with a **platinum catalyst**.

Methanol and other hydrogen-rich fuels

Hydrogen is hard to store (a gas with low energy per volume) and flammable. A liquid **hydrogen-rich fuel** such as methanol is easier to handle:

- Anode (acid): $CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$
- Cathode: $1½O_2 + 6H^+ + 6e^- \rightarrow 3H_2O$
- Overall: $\mathbf{CH_3OH + 1½O_2 \rightarrow CO_2 + 2H_2O}$

Advantages and disadvantages

Advantages	Disadvantages
H_2 cell: only **water** made; no CO_2 at the point of use (combustion engines give CO_2, CO, NO_x)	making H_2 needs energy, often from fossil fuels (electrolysis or from methane)
more **efficient** than burning the fuel in an engine	H_2 is **flammable**; storing and transporting it is difficult (high pressure or low temperature)
quiet, few moving parts	expensive platinum catalyst; fuel cells have a limited lifetime

MARK-SCHEME WORDING

- How it works: fuel cells "use the energy released on the reaction of a fuel / hydrogen with oxygen to generate a voltage" (WCH15 Jun 2024).
- Advantage: "only water is produced, whereas CO_2 is produced in an internal combustion engine". Disadvantage: "production / transport / storage of hydrogen is difficult". Another fuel: "e.g. methanol (as it is hydrogen-rich)".
- Methanol oxidation: "$CH_3OH + H_2O \rightarrow 6H^+ + CO_2 + 6e^-$". DO NOT ACCEPT the formula CH_2OH (WCH15 Oct 2024).

Often asked as: 6-mark "explain how a hydrogen fuel cell works, with an advantage, a disadvantage and another fuel"; MCQs on electrode half-equations.

COMMON MISTAKE

Forgetting to explain **how** the cell works (the first indicative point was the least seen, WCH15 Jun 2024 examiner report). Not answering the last part of the question ("name another fuel").

Treating "flammable" and "explosive" as the same thing. Writing methanol as CH_2OH.

SPEC 16.19 Electrode reactions in a hydrogen–oxygen fuel cell

WHAT THIS SPEC POINT ASKS

Know the electrode reactions in a hydrogen–oxygen fuel cell with an acidic and with an alkaline electrolyte.

	Acidic electrolyte	**Alkaline** electrolyte (e.g. KOH)
Ion carrying charge	H^+ (anode → cathode)	OH^- (cathode → anode)
Anode (−), oxidation	$H_2 \rightarrow 2H^+ + 2e^-$	$H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$
Cathode (+), reduction	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$
E° values (as reductions)	$2H^+/H_2$ 0.00 V; O_2/H_2O +1.23 V	H_2O/H_2, OH^- −0.83 V; O_2/OH^- +0.40 V
E°cell	1.23 − 0.00 = **+1.23 V**	0.40 − (−0.83) = **+1.23 V**
Overall	**$2H_2 + O_2 \rightarrow 2H_2O$**	**$2H_2 + O_2 \rightarrow 2H_2O$**

- The overall reaction and E°cell are the **same** in acid and in alkali: only the ions in the half-equations differ.
- Check which medium the question gives: use **H^+** in acid and **OH^-** in alkali, never a mixture.

MARK-SCHEME WORDING

- Acid: anode "$H_2(g) \rightleftharpoons 2H^+(aq) + 2e^-$"; cathode "$O_2(g) + 4H^+(aq) + 4e^- \rightleftharpoons 2H_2O(l)$" (WCH15 Jun 2024).
- Alkaline: anode "$H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$"; cathode "$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$" (WCH15 Jan 2021; Jun 2025).
- Overall "$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$", E° = +1.23 V (WCH15 Jan 2024).

Often asked as: MCQs picking the anode or cathode half-equation in acid or alkali; writing the reduction half-equation for an unfamiliar cell.

COMMON MISTAKE

Mixing acid and alkaline half-equations, or balancing with OH^- when H^+ and H_2O were specified (WCH15 Oct 2024 examiner report).

Mixing up anode and cathode. In any cell, the **an**ode is where **oxidation** happens (both start with a vowel); in a fuel cell it is the **negative** terminal.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Other cells built from the same ideas: zinc–air ($Zn + 2OH^- \rightarrow Zn(OH)_2 + 2e^-$, WCH15 Jan 2022) and the alkaline Zn/MnO_2 cell ($2MnO_2 + H_2O + 2e^- \rightarrow Mn_2O_3 + 2OH^-$, WCH15 Jan 2023). A Nernst-type expression applied to a fuel cell (WCH15 Oct 2024).

Quick checks

QUICK CHECK

What is $E^{\ominus}_{cell}$ for Mg(s) | Mg^{2+}(aq) || Cu^{2+}(aq) | Cu(s)?

- −2.71 V
- −2.03 V
- +2.71 V
- +1.71 V

$E^{\ominus}_{cell} = E_{right} - E_{left} = +0.34 - (-2.37) = +2.71$ V. −2.03 V comes from adding the two values instead of subtracting; −2.71 V from subtracting the wrong way round.

QUICK CHECK

Why is the salt bridge usually soaked in potassium nitrate rather than potassium chloride?

- KNO_3 conducts electrons better
- Chloride ions could react with some solutions, for example precipitating Ag^+ as AgCl
- KCl would react with the voltmeter
- Nitrate ions take part in the cell reaction

The salt bridge carries ions, not electrons, and its ions must not react with either half-cell.

QUICK CHECK

Which statement about the standard hydrogen electrode is correct?

- it uses hydrogen at 1 atm and 1.00 mol dm^{-3} NaOH
- its potential is measured as 0 V against a calomel electrode
- it uses H_2 at 100 kPa, 1.00 mol dm^{-3} H^+ and a platinum electrode at 298 K
- the platinum is there to react with the hydrogen

Its potential is defined as 0 V, not measured. The platinum black is an inert, conducting surface that catalyses the H_2/H^+ equilibrium.

QUICK CHECK

Using the table, which species could oxidise Br^- to Br_2 under standard conditions?

- Fe^{3+}
- I_2
- Cu^{2+}
- Cl_2

The oxidising agent must have a more positive $E^{\ominus}$ than Br_2/Br^- (+1.09 V). Cl_2/Cl^- is +1.36 V, giving $E^{\ominus}_{cell}$ = +0.27 V; the others give negative values.

QUICK CHECK

$E^{\ominus}_{cell}$ for a reaction is +0.45 V, but no reaction is seen. What is the best explanation?

- the activation energy is too high, so the reaction is too slow
- the reaction is not feasible
- $E^{\ominus}_{cell}$ should be negative for a reaction to occur
- K is less than 1

A positive $E^{\ominus}_{cell}$ means the reaction is feasible and K > 1, but it says nothing about the rate.

QUICK CHECK

Why is hydrochloric acid not used to acidify potassium manganate(VII) in titrations?

- it is too weak to supply enough H^+
- it reduces Fe^{3+} back to Fe^{2+}
- MnO_4^- oxidises Cl^- to Cl_2, so more manganate(VII) is used
- it forms a precipitate with Mn^{2+}

$E^\ominus(MnO_4^-/Mn^{2+})$ = +1.51 V is more positive than $E^\ominus(Cl_2/Cl^-)$ = +1.36 V, so the titre would be too large and toxic chlorine would form.

QUICK CHECK

0.100 g of potassium iodate(V), KIO_3 (M_r 214), is added to excess acidified KI. What volume of 0.100 mol dm^{-3} thiosulfate reacts with the iodine formed?

- 4.67 cm^3
- 28.0 cm^3
- 14.0 cm^3
- 9.35 cm^3

Moles of IO_3^- = 0.100 ÷ 214 = 4.67 × 10^{-4}. Each makes $3I_2$, which need 6 $S_2O_3^{2-}$: 2.80 × 10^{-3} mol, which is 28.0 cm^3. 14.0 cm^3 forgets that each I_2 needs two thiosulfate ions.

QUICK CHECK

In an iodine–thiosulfate titration, when should the starch be added?

- at the start, so the colour change is easier to see
- near the end point, when the iodine colour has faded to pale yellow
- after the end point, to confirm it
- it is not needed because thiosulfate is coloured

Added early, the starch–iodine complex forms in large amounts and does not release its iodine readily, giving an inaccurate end point.